

**771765 Research and Application in AI and Data Science:
Research Proposal Assignment**

(This assignment is worth 50% of the total marks for this module)

This assignment should be submitted **via turnitin on CANVAS** by
Thursday 30th April 2026 at 14:00

Maximum Word Count for the written part of this assignment = 3000 words.

Context

This assignment is **critical** to trimester 3 as it will define your initial research project scope and work plan. To produce this assignment, students must carefully think about what project they wish to undertake in the third trimester, negotiate with potential supervisors and/or external companies, document the goals and general approaches, and ensure appropriate ethical considerations have been met prior to trimester 3 commencing. Please review the recorded lecture series for this module to help.

Students are permitted to give one (1) draft of the assignment to their supervisor for informal feedback prior to submitting the final assignment. The primary purpose of this draft is to ensure that the proposal is at the correct level for an MSc project.

Requirements

The research proposal must:

- (a) **State the main aims or objectives** of the research to be undertaken in trimester 3. These aims or objectives should be succinct (no more than a few sentences).
- (b) **Spell out the background** to the research that is being proposed. This must include citations of the literature surrounding the project to give its context.
- (c) **Detail what the research will do, along with the anticipated outcomes.** This will likely include the methods (or the anticipated methods – we acknowledge this may change in trimester 3) and a clear statement of what the outcomes will be.
- (d) **Include a timeline (Gantt Chart)** of the various tasks that will need to be undertaken during the research.
- (e) **Include ethical clearance forms**

Note that the word count includes only (a), (b), (c), and (d).

Please upload one (1) document to the research proposal submission area which contain points (a) – (d). The ethical clearance form is submitted to a separate area.

Grading

The following grading rubric will be applied to submissions. Please note that there are two “hurdle criteria” which must be passed in order to score any grades for this assessment item, as below. The total number of marks available for this assignment is 100.

Please upload:

- (i) Your written research proposal to the assignment, including any figures.

- (ii) Your ethical clearance form(s).
- (iii) A proposed Gantt chart of the timeline for your project.

Hurdle Criterion Number 1: MSc Level Research.

0 marks will be awarded for the entire submission if the proposal is not at the required level for MSc research. If you have a self-designed project, please supply an early draft to Dr Cashmore, or discuss directly with them, to ensure that your proposal is at the right level. If your project is from the list, please send this to the relevant supervisor.

Hurdle Criterion Number 2: Ethics Clearance.

0 marks will be awarded for the entire submission in the absence of an ethical clearance form. If in any doubt what ethical clearance is required, please talk to Dr Cashmore as a matter of urgency.

Further Grading Criteria**Research project background and referencing:**

This section should include an introduction to your project topic. Here you can present the relevant background research and include citations of the literature surrounding the project to give its context. When you are discussing previous work, you should aim to summarise their findings along with the methods used in order to justify your approach in the next section. Ensure your citations are formatted correctly (Harvard or Vancouver) and are relevant to the topic.

Research description and anticipated outcomes:

This section should present the methodology. Present a clear statement of methods with an explanation of why they are being used and justify this by connecting to the previous work described in the background section. You should also include the anticipated outcomes of the project and justify them by connecting to the background. What do you expect to find out and why? An outcome of "this work will explore trends in..." is not enough.

Timeline:

The timeline should outline the various tasks that must be completed throughout the duration of the project. These tasks should be broken down into more detailed steps that you have carefully thought through and assigned a reasonable time frame to complete. Risk mitigation should also be considered here; are there any areas that may present issues? If so, then how will you account for this in the timeline?

Please see the below rubric for guidance.

Please note that the total word limit for the entire document is 3000 words.

Research project background and referencing (40 marks)					
	1	2	3	4	5
Background (20)	An attempt at the background is made, but it is significantly under researched.	The background is Incomplete and missing key concepts.	The background is sufficient to plan and complete the project and justifies the aims.	The background is more complete and covers a greater depth.	The level of background research is at a professional level.
References (10)	Incorrectly written references whether in text or bibliography .	Some references present but clearly thrown in with little thought.	Referencing that covers a basic level.	More depth applied to the use of the references.	Highly relevant references as needed. No over referencing of material.
Relevancy (10)	The material presented is not relevant to the project as a whole.	The material presented is off-topic, but close in subject matter. Misunderstanding of the brief.	The material presented is on-topic.	The material presents some basic connections between the background and the topic.	In depth background with connection to the research question.
Research description and anticipated outcomes (40 marks)					
	1	2	3	4	5
Research Description (20)	An attempt of a research description is made. One sentence of plan, for example, I will do X, I will do Y,...	A simple research description is given which lacks sufficient detail to complete the project.	A research description that is sufficient to complete the project, but minimum detail is provided.	A research description with some depth, but not at a professional level.	Fully explored description of planned work with thought put into each aspect.
Outcomes (10)	Extremely basic outcomes are presented, e.g. "I will make an AI".	A simple outcome is provided which is missing detail.	An outcome is given with minor details.	The outcomes are justified and have some connections to the background.	A fully connected description of outcomes which provide further insight, at a professional level of understanding.

Justification for the research plan (10)	No justification of work past what the student wants to do.	Some justification is given, but not enough to justify the work.	Basic levels of justification, enough to justify the research direction.	Further justification, but not complete or missing citations.	Fully justified, with strong connections made to the background and citations used.
Timeline (20 marks)					
	1	2	3	4	5
Timescales (10)	An attempt at timescales is made, but it is inappropriate for the research plan.	The timescales appear unreasonable considering the work to be undertaken.	The timescales are sensible, but not justified.	The timescales are well thought out, with minor justification.	The timescales are fully justified, e.g., time for running code.
Tasks (5)	Key steps are missing.	Steps are poorly written or unclear.	The tasks outline the basic steps to finish the project.	The tasks show steps that are thought out and clear.	The tasks show steps along with a discussion.
Risks to the timeline (5)	The bare minimum of risk analysis is provided.	Some risks are acknowledged, but not relevant to the timeline.	An inclusion of risk analysis that is relevant.	The risks are planned out, but incomplete and missing contingencies.	Fully planned, with contingencies included.